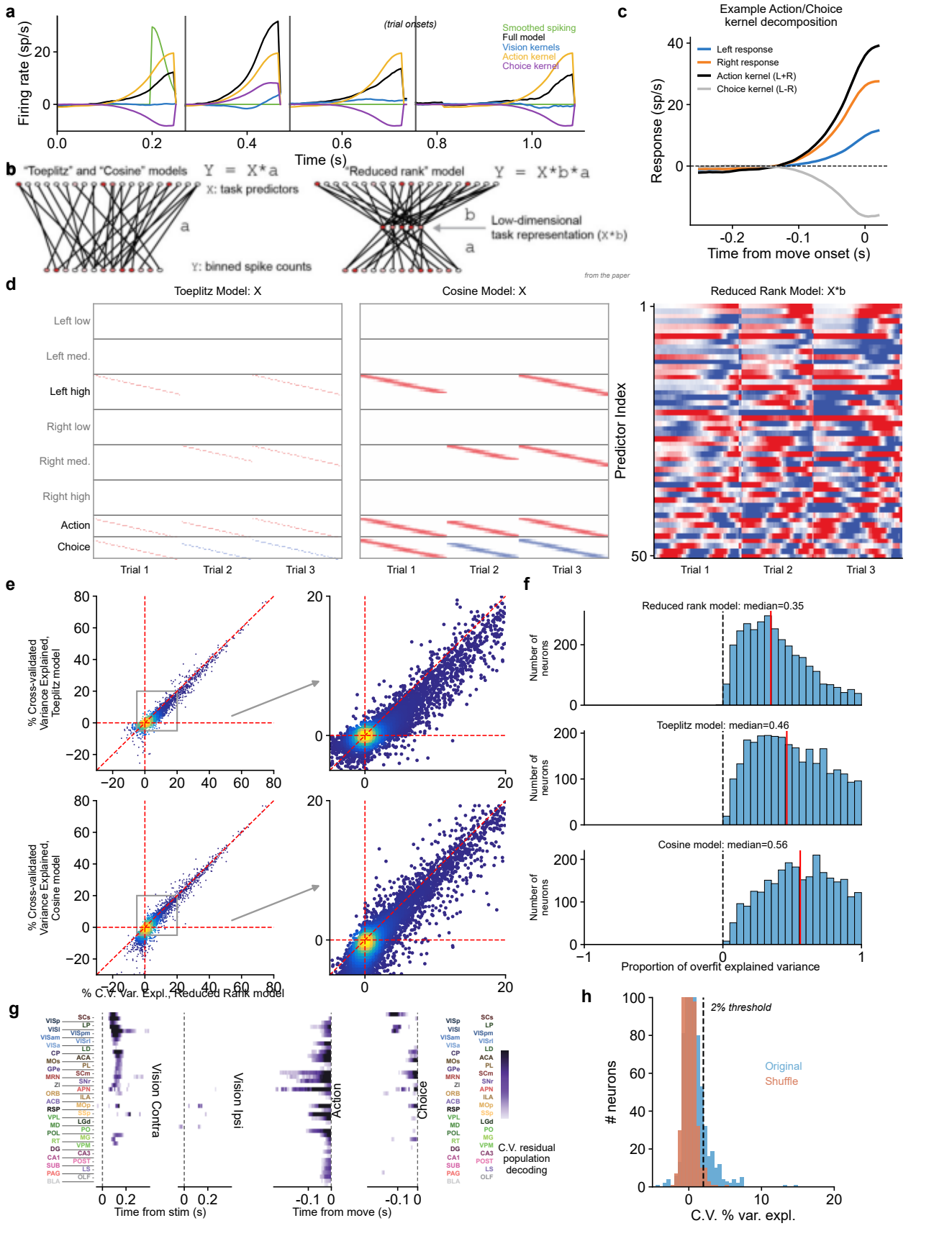

a
Firing rate (sp/s)
(trial onsets)
Smoothed spiking
Full model
Vision kernels
Action kernel
Choice kernel
Time (s)
b
"Toeplitz" and "Cosine" models Y = X*a
X: task predictors
a
Y: binned spike counts
"Reduced rank" model Y = X*b*a
b
a
Low-dimensional task representation (X*b)
from the paper
c
Example Action/Choice kernel decomposition
Left response
Right response
Action kernel (L+R)
Choice kernel (L-R)
Response (sp/s)
Time from move onset (s)
d
Toeplitz Model: X
Cosine Model: X
Reduced Rank Model: X*b
Left low
Left med.
Left high
Right low
Right med.
Right high
Action
Choice
Predictor Index
Trial 1
Trial 2
Trial 3
e
% Cross-validated Variance Explained, Toeplitz model
% Cross-validated Variance Explained, Cosine model
% C.V. Var. Expl., Reduced Rank model
f
Reduced rank model: median=0.35
Toeplitz model: median=0.46
Cosine model: median=0.56
Number of neurons
Proportion of overfit explained variance
g
Vision Contra
Vision Ipsi
Action
Choice
Time from stim (s)
Time from move (s)
C.V. residual population decoding
h
2% threshold
Original
Shuffle
neurons
C.V. % var. expl.